

Forces: Push, Pull and Motion

Name: _____

Date: _____

Year 4 Science — Physical Sciences

ACARA v9.0: AC9S4U03

Learning Intention

I am learning to investigate the effect of frictional, gravitational and magnetic forces on the motion of objects.

Success Criteria

- I can identify friction, gravity and magnetism.
- I can explain how each force affects motion.
- I can predict motion from forces acting on it.

3 Forces, 3 Effects

Friction slows things down when surfaces rub together. Gravity pulls objects towards the Earth. Magnetism pulls or pushes certain metal objects without touching them.

Worked Example

Why does a bike slow down if you stop pedalling?
Friction between the tyres and the road (and air resistance) acts against the motion, slowing the bike down.

Part A — Warm Up

Circle the force that pulls a dropped ball to the ground: **friction** **gravity** **magnetism**

True or False: Magnetism can pull objects without touching them. _____

Name one surface that creates a lot of friction (hard to slide on). _____

Part B — Core Skills

Identify the force at work: A fridge magnet sticks to a metal fridge door. Force:

Explain why a ball rolls further on smooth tiles than on thick carpet.

List 2 objects that would be attracted to a magnet.

1. _____ 2. _____

Why do skydivers eventually fall at a steady speed instead of getting faster and faster forever?

Word Bank

force • friction • gravity • magnetism • motion • attract • repel • surface • air resistance

Working Space — Draw an Example of Friction, Gravity and Magnetism

Forces: Push, Pull and Motion — Page 2

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Part C — Challenge: The Skate Park Investigation

Ollie is testing his skateboard at the local skate park in Geelong. He notices the skateboard rolls much faster and further on the smooth concrete bowl than on the gravel path near the entrance.

Q8. Identify which force is mainly responsible for slowing Ollie's skateboard down faster on the gravel path.

Q9. Explain, using force language, why the skateboard's wheels eventually stop even on the smooth concrete if Ollie doesn't push again.

Q10. Ollie jumps off a ramp into the air. Explain which force brings him back down to the ground, and describe what would happen (in theory) if that force suddenly disappeared.

Answer Key — Part A

A1: gravity

A2: True — magnetism is a non-contact force that can act at a small distance.

A3: Carpet, sand, gravel, or a rough road (accept any reasonable rough surface).

Answer Key — Part B

B1: Magnetism.

B2: Smooth tiles create less friction against the ball than thick carpet, so the ball keeps moving further before friction slows it to a stop.

B3: Any 2 reasonable magnetic objects (e.g. paperclip, nail, fridge, tin can) — teacher to check.

B4: As skydivers fall faster, air resistance (a type of friction with the air) increases until it balances the pull of gravity, so their speed becomes steady instead of continuously increasing.

Answer Key — Part C

C8: Friction — the rough gravel surface creates more friction against the wheels than the smooth concrete.

C9: Even on smooth concrete, a small amount of friction between the wheels and ground (and air resistance) gradually acts against the motion, slowing the skateboard until it stops.

C10: Gravity pulls Ollie back down to the ground; if gravity suddenly disappeared, Ollie would keep floating upward/outward in the direction he was moving instead of coming back down, since nothing would pull him toward the Earth.

I Can Checklist

☐ I can identify friction, gravity and magnetism.

☐ I can predict how objects will move.

☐ I can explain how each force affects motion.